DES Pune University, Pune
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Subject: FCP		
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Title:	**To write a C++ program to perform file operations**	
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Contents

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Theory(Logic)

This C++ program demonstrates basic file handling using the fstream library. It begins by creating an output file stream using ofstream to open or create a file named sample.txt. The user enters text, which is read line by line using getline(). Each entered line is written to the file, and this continues until the user inputs "-1", which acts as a sentinel value to terminate the loop. After completing the writing process, the output file stream is closed. Next, the program opens the same file using an input file stream, ifstream, to read its contents. Using another loop, it reads each line from the file using getline() and displays the line on the screen using cout. This continues until the end of the file is reached. Finally, the input file stream is closed, and the program ends. This approach ensures that data entered by the user is stored in a file and then retrieved and displayed, demonstrating both writing to and reading from a file in C++.

Pseudo Code

- **Step 1: Include input-output library**
- **Step 2: Use standard namespace**
- **Step 3:Declare file stream for output (Myfile)**
- **Step 4:Declare string variable (line)**
- **Step 5:Open file "sample.txt" for writing using Myfile**
- **Step 6:While file is open**
    - **Read line from user input**
    - **If line is equal to "-1"**
    - **Exit the loop**
    - **Write the line into the file with a newline**
- **Step 7:End While**
- **Step 8:Close the output file**
- **Step 9:Declare file stream for input (MyReadFile)**
- **Step 10:Open file "sample.txt" for reading using MyReadFile**
- **Step 11:While there is a line to read from the file**
    - **Read line from file**
    - **Display line on the console**
- **Step 12:End While**
- **Step 13:Close the input file**
- **Step 14: End main function**

Program Code

```
#include<iostream>
#include<fstream>
using namespace std;

int main()
{
   ofstream Myfile;
   string line;
   Myfile.open("sample.txt");
   while(Myfile)
   {
      getline(cin,line);
      if(line=="-1")
      break;
      Myfile<<line<<endl;
   }
   Myfile.close();

   ifstream MyReadFile;
   MyReadFile.open("sample.txt");
   while(getline(MyReadFile,line))
   {
      cout<<line<<endl;
   }
   MyReadFile.close();
   return 0;
}
```

Output

```
hello this is my code
-1
hello this is my code
```

```
main.cpp    sample.txt
1  hello this is my code
2
```

Conclusion

Code is run successfully, required output is received.